

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

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Abstract

We introduce a new language representation model called **BERT**, which stands for **Bidirectional Encoder Representations from Transformers**. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of tasks, such as question answering and language inference, without substantial task-specific architecture modifications.

BERT is conceptually simple and empirically powerful. It obtains new state-of-the-art results on eleven natural language processing tasks, including pushing the GLUE score to 80.5% (7.7% point absolute improvement), MultiNLI accuracy to 86.7% (4.6% absolute improvement), SQuAD v1.1 question answering Test F1 to 93.2 (1.5 point absolute improvement) and SQuAD v2.0 Test F1 to 83.1 (5.1 point absolute improvement).

1 Introduction

Language model pre-training has been shown to be effective for improving many natural language processing tasks (Dai and Le, 2015; Peters et al., 2018a; Radford et al., 2018; Howard and Ruder, 2018). These include sentence-level tasks such as natural language inference (Bowman et al., 2015; Williams et al., 2018) and paraphrasing (Dolan and Brockett, 2005), which aim to predict the relationships between sentences by analyzing them holistically, as well as token-level tasks such as named entity recognition and question answering, where models are required to produce fine-grained output at the token level (Tjong Kim Sang and De Meulder, 2003; Rajpurkar et al., 2016).

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the same objective function during pre-training, where they use unidirectional language models to learn general language representations.

We argue that current techniques restrict the power of the pre-trained representations, especially for the fine-tuning approaches. The major limitation is that standard language models are unidirectional, and this limits the choice of architectures that can be used during pre-training. For example, in OpenAI GPT, the authors use a left-to-right architecture, where every token can only attend to previous tokens in the self-attention layers of the Transformer (Vaswani et al., 2017). Such restrictions are sub-optimal for sentence-level tasks, and could be very harmful when applying fine-tuning based approaches to token-level tasks such as question answering, where it is crucial to incorporate context from both directions.

In this paper, we improve the fine-tuning based approaches by proposing BERT: **Bidirectional Encoder Representations from Transformers**. BERT alleviates the previously mentioned unidirectionality constraint by using a “masked language model” (MLM) pre-training objective, inspired by the Cloze task (Taylor, 1953). The masked language model randomly masks some of the tokens from the input, and the objective is to predict the original vocabulary id of the masked

Language Models are Few-Shot Learners

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Abstract

We demonstrate that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even becoming competitive with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks. We also identify some datasets where GPT-3’s few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora.

1 Introduction

NLP has shifted from learning task-specific representations and designing task-specific architectures to using task-agnostic pre-training and task-agnostic architectures. This shift has led to substantial progress on many challenging NLP tasks such as reading comprehension, question answering, textual entailment, among others. Even though the architecture and initial representations are now task-agnostic, a final task-specific step remains: fine-tuning on a large dataset of examples to adapt a task-agnostic model to perform a desired task.

Recent work [RWC⁺19] suggested this final step may not be necessary. [RWC⁺19] demonstrated that a single pretrained language model can be zero-shot transferred to perform standard NLP tasks

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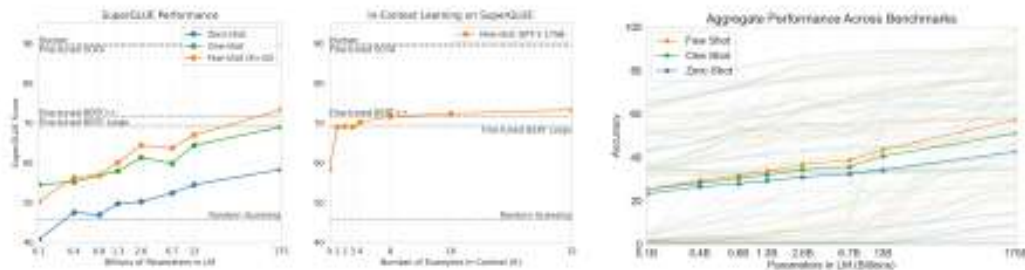


Figure 1.1: Performance on SuperGLUE increases with model size. A value of $K = 32$ means that our model was shown 32 examples per task, for 256 examples total divided across the 8 tasks in SuperGLUE. We report GPT-3 values on the dev set, so our numbers are not directly comparable to the dotted reference lines (our test set results are in the appendix). The BERT-Large reference model was fine-tuned on the SuperGLUE training set (125K examples), whereas BERT++ was first fine-tuned on MultiNLI (392K examples) and SWAG (113K examples) before further fine-tuning on the SuperGLUE training set (for a total of 630K fine-tuning examples).

Performance on SuperGLUE increases with number of examples in context. We find the difference in performance between the BERT-Large and BERT++ to be roughly equivalent to the difference between GPT-3 with one example per context versus eight examples per context.

Aggregate performance for all 42 accuracy-denominated benchmarks. While zero-shot performance improves steadily with model size, few-shot performance increases more rapidly, demonstrating that larger models are more proficient at in-context learning.

without the need for finetuning on a dataset of training examples. While this work was a promising proof of concept, the best case performance only matched some supervised baselines on a single dataset. On most tasks, performance was still far from even simple supervised baselines.

However [RWC⁺19] also showed a potential way forward. The work observed relatively consistent log-linear trends in performance on both transfer tasks and language modeling loss across one an order of magnitude of scaling. [KMH⁺20] then conducted a much more rigorous study of the scaling behavior of log loss and confirmed smooth scaling trends. In this work, we empirically test whether scaling continues to improve performance by extrapolating the previously identified phenomena another two orders of magnitude. We train a 175 billion parameter autoregressive language model, which we call GPT-3, and measure its transfer learning abilities.

As part of this investigation, we also clarify and systematize the approach introduced in [RWC⁺19]. While [RWC⁺19] describe their work as “zero-shot task transfer” they sometimes provide examples of the relevant task in the context. Due to the use of what are effectively training examples, these cases are better described as “one-shot” or “few-shot” transfer. We study these one-shot and few-shot settings in detail comparing them with the zero-shot setting which only uses a natural language description or invocation of the task to be performed. Our findings are summarized in Figure 1.1. We observe that one- and few-shot performance is often much higher than true zero-shot performance leading us to suggest that language models can also be understood as meta-learners where slow outer-loop gradient descent based learning is combined with fast “in-context” learning implemented within the context activations of the model.

Broadly, on NLP tasks GPT-3 achieves promising results in the zero- and one-shot settings, and in the few-shot setting is sometimes competitive with or even occasionally surpasses state-of-the-art (despite state-of-the-art being held by fine-tuned models). For example, GPT-3 achieves 81.5 F1 on CoQA in the zero-shot setting, 84.0 F1 on CoQA in the one-shot setting, and 85.0 F1 in the few-shot setting. Similarly, GPT-3 achieves 64.3% accuracy on TriviaQA in the zero-shot setting, 68.0% in the one-shot setting, and 71.2% in the few-shot setting, the last of which is state-of-the-art relative to fine-tuned models operating in the same closed-book setting.

We additionally train a series of smaller models (ranging from 125 million parameters to 13 billion parameters) in order to compare their performance to GPT-3 in the zero-, one- and few-shot settings. In general, we find relatively smooth scaling for most tasks with model capacity in all three settings; one notable pattern is that the gap between zero-, one-, and few-shot performance often grows with model capacity, perhaps suggesting that larger models are more proficient meta-learners.

LEGAL-BERT: The Muppets straight out of Law School

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Abstract

BERT has achieved impressive performance in several NLP tasks. However, there has been limited investigation on its adaptation guidelines in specialised domains. Here we focus on the legal domain, where we explore several approaches for applying BERT models to downstream legal tasks, evaluating on multiple datasets. Our findings indicate that the previous guidelines for pre-training and fine-tuning, often blindly followed, do not always generalize well in the legal domain. Thus we propose a systematic investigation of the available strategies when applying BERT in specialised domains. These are: (a) use the original BERT out of the box, (b) adapt BERT by additional pre-training on domain-specific corpora, and (c) pre-train BERT from scratch on domain-specific corpora. We also propose a broader hyper-parameter search space when fine-tuning for downstream tasks and we release LEGAL-BERT, a family of BERT models intended to assist legal NLP research, computational law, and legal technology applications.

1 Introduction

Pre-trained language models based on Transformers (Vaswani et al., 2017), such as BERT (Devlin et al., 2019) and its variants (Liu et al., 2019; Yang et al., 2019; Lan et al., 2019), have achieved state-of-the-art results in several downstream NLP tasks on generic benchmark datasets, such as GLUE (Wang et al., 2018), SQUAD (Rajpurkar et al., 2016), and RACE (Lai et al., 2017).

Typically, transfer learning with language models requires a computationally heavy step where the language model is pre-trained on a large corpus and a less expensive step where the model is fine-tuned for downstream tasks. When using BERT, the first step can be omitted as the pre-trained models are publicly available. Being pre-trained on



Figure 1: The three alternatives when employing BERT for NLP tasks in specialised domains: (a) use BERT out of the box, (b) further pre-train BERT (FP), and (c) pre-train BERT from scratch (SC). All strategies have a final fine-tuning step.

generic corpora (e.g., Wikipedia, Children’s Books, etc.) BERT has been reported to under-perform in specialised domains, such as biomedical or scientific text (Lee et al., 2019; Beltagy et al., 2019). To overcome this limitation there are two possible strategies; either further pre-train (FP) BERT on domain specific corpora, or pre-train BERT from scratch (SC) on domain specific corpora. Consequently, to employ BERT in specialised domains one may consider three alternative strategies before fine-tuning for the downstream task (Figure 1): (a) use BERT out of the box, (b) further pre-train (FP) BERT on domain-specific corpora, and (c) pre-train BERT from scratch (SC) on domain specific corpora with a new vocabulary of sub-word units.

In this paper, we systematically explore strategies (a)–(c) in the legal domain, where BERT adaptation has yet to be explored. As with other specialised domains, legal text (e.g., laws, court pleadings, contracts) has distinct characteristics compared to generic corpora, such as specialised vocabulary, particularly formal syntax, semantics based on extensive domain-specific knowledge etc., to the extent that legal language is often classified as a ‘sublanguage’ (Tiersma, 1999; Williams, 2007; Haigh, 2018). Note, however, that our work contributes more broadly towards a better understanding of domain adaptation for specialised domains. Our key findings are: (i) Further pre-training (FP) or pre-training BERT from scratch (SC) on domain-

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 13].

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

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Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t , as a function of the previous hidden state h_{t-1} and the input for position t . This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [28].

To the best of our knowledge, however, the Transformer is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence-aligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [14, 15] and [8].

3 Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 29]. Here, the encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, \dots, y_m)$ of symbols one element at a time. At each step the model is auto-regressive [9], consuming the previously generated symbols as additional input when generating the next.

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder, shown in the left and right halves of Figure 1, respectively.

3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-